Exercises

~~Exercise~~ 4 Inner Join SQL

Question 1
Inner Join

① Syntax

```
Select  Column 1
        Column 2,
        Column 3,
FROM table 1 AS A
INNER JOIN table 2 AS B
ON A. common_column = B. common_column;
```

```
Select Employee ID, first Name, Last Name, Department, Salary
,Project ID, Project_Name, Budget, Status,

FROM employees AS A
INNER JOIN Projects AS B
ON A. Employee_ID = B. Employee_ID;
```

Outcome

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Develop	100000	Complete
2	Alice	Smith	HR	60000	102	Employee Training	80000	Ongoing
3	Bob	Johnson	Finance	75000	103	Financial Audit	90000	Pending
4	David	Brown	IT	72000	104	Cyber Security	75000	ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Complete
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

(2) LEFT JOIN.

```
SELECT EmployeeID, first Name, Department Salary
       ProjectID, Project Name, Budget, Status
FROM Employees AS A
LEFT JOIN ~~Department~~ Projects AS B
ON AEmployee ID = B Employee_ID ;
```

LastName	Employee ID	first Name	Department	Salary	Project ID	Project Name	Budget
Doe	1	John	~~IT~~	70000	101	AI Dev	100000
Smith	2	Alice	HR	60000	102	Employee Training	50000
Johnson	3	Bob	Finance	75000	104	Financial Analysis	90000
Brown	4	David	IT	72000	103	Cybersecurity	75000
Wilson	5	Emma	Sales	65000	105	Marketing Expansion	65000
Clark	6	Michael	Finance	80000	106	Risk Management	80000

Status
Complete
Ongoing
ongoing
Pending
Complete
Pending

③ Right JOIN

```
SELECT Employee ID, First Name, Last Name, Department
       Salary
FROM Projects AS A.
RIGHT JOIN Employees AS B
ON A. Employee_ID = B Employee_ID
```

EmployeeID	First Name	Last Name	Department	Salary	Project ID
1	John	Doe	IT	70000	
2	Alice	Smith	HR	60000	
3	Bob	Johnson	finance	75000	
5	Emma	Wilson	Sales	65000	
6	Michael	Clark	finance	80000	
1	David	Brown	IT	72000	

④ Full Outer Join

```
SELECT Employee_ID, first_Name, Department, Salary
       , City, Age, Project_ID, Project Name, Budget
       Start date, Status
FROM Employees AS A
FULL IN OUTERJOIN Projects AS B
ON A Employee_ID = B. Employee_ID;
```

Outcome

Employee ID	First Name	Last Name	Dep	Salary	City	Age	Project ID	Project Name
Null	John	Doe	IT	70000	New York	35	101	AI Develop
2	Alice	Smith	HR	60000	Los Angeles	40	102	Employee training
3	Bob	Johnson	Finance	75000	Toronto	45	104	Finance Analysis
4	Emma	Wilson	Sales	65000	Sydney	28	105	Market Expansion
5	Michael	Clark	Finance	80000	New York			
6								
Null	David	Brown	IT	~~80000~~ 72000	~~New York~~ London	45	10	Cyber Security

Budget	Start Date	Status
75000	2024-02-15	Ongoing
100 000	2024-01-10	Completed
50000	2024-02-15	Ongoing
95000	2024-04-12	ongoing
90 000	2024-05-20	Complete

-2 Union & Union All

Select* City, AS Location
~~Status~~ Status,
FROM Employee_table;

Employee ID	Location	Status.
1	Newyork	complete
2	Los Angeles	Ongoing
3	Toronto	Pending
4	London	Ongoing
5	Sydney	complete
6	New York	Pending

⑧ SELECT* Employee_ID, first_Name, Last_Name, Department, Salary
FROM Employee
WHERE Department = finance OR IT

Employee_ID	First Name	Last Name	Department	Salary
1	John	Doe	IT	70000
4	David	Brown	IT	72000
3	Bob	Johnson	Finance	75000
6	Michael	Clark	Finance	80000

⑨ Select* Project ID, Project Name, Budget, Status
FROM ~~Employees~~ Projects
WHERE Status = NOT complete;

Project ID	Project Name	Budget	Status
102	Employee Training	50000	Ongoing
103	CyberSecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

⑩ Select*
FROM Project
WHERE ~~Project~~ Budget > 70000 AND Status NOT complete;

Project ID	Project Name	Budget	Status
~~100~~	~~AI Development~~	~~100000~~	
~~102~~	~~Employee Training~~		
103	CyberSecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

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```
SELECT city, ~~As New York~~
       Salary
FROM Employees
GROUP BY City(Toronto, New York)
ORDER BY Salary DESC;
```

Descending

Employee ID	First Name	Last Name	Department	Salary	City
				~~New~~	New York
1	John	Doe	IT	120000	New York
3	Bob	Johnson	Finance	75000	Toronto
6	Michael	Clark	Finance	80000	New York

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```
SELECT Employee ID,
       First Name,
       Last Name,
       Department,
       Salary
FROM Employee
WHERE Salary = highest
LIMIT 3;
```

Employee ID	First Name	Last Name	Department	Salary
~~2~~	David	Brown	IT	72000
3	Bob	Johnson	Finance	75000
6	Michael	Clark	Finance	80000

④

13 Aggregate functions with Group BY & Having

```
Select SUM Department As total salary
FROM Employee
ORDER BY DESC;
```

Department	Total Salary
IT	142 000
HR	60000
Finance	155000
Sales	65000

(14) Select *
SUM (total Salary AS Average Salary)
FROM 'employee'
Group By ~~Salary average~~ City
WHERE Average Salary > 65000

Salary combined / Number of empl

City	Average Salary > 65000
New York	75000
Toronto	75000
London	72000

15. COUNT Employee_ID,
Department
FROM 'Employee'
GROUP BY Employee Count = > 1 ;

Employee Count	Department
2	IT
2	Finance

(16) SELECT Project Count,
Status
FROM Project
WHERE Status ~~AS~~ Project ID (2);

Status	Project Count
Complete	2
Ongoing	2
Pending	2

(17) SELECT ~~SUM~~ First Name, Last Name, Project ID, Budget
sum (Budget) AS total Budget)
FROM ~~Bud~~ Project
~~WHE~~ Having = Budget 7150000;

Project ID	Name	Last Name	Budget
101	John	Doe	Null
102	Alice	Smith	Null
103	David	Brown	Null
104	Bob	Johnson	Null
105	Emma	Wilson	Null
106	Michael	Clark	Null